Smart Programming : YouTube Channel

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JDBC (PART 17)

Connection Pooling

Different ways to provide Connection pooling

1. Default connection pooling provided by database vendors

   (no need to use external jar files)

2. Third party connection pooling mechanism

   (we have to use external jar files)

   a. Apache Commons DBCP (Database Connection Pooling)

   b. Tomcat JDBC

   c. C3P0

   d. Proxool

   e. BoneCP

f. Hikary

3. Application server provided connection pooling

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1. Default connection pooling provided by database vendors

(no need to use external jar files)

➢ **Steps for Connection pooling**

1. Create DataSource object

for MySQL - MySQLDataSource

for Oracle - OracleDataSource

2. Set connection properties to the DataSource

Methods used are

-> setURL(Stirng driver_url)

-> setUser(String username)

-> setPassword(String password)

3. Get connection object form DataSource

-> getConnection()

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2. Third party connection pooling mechanism

(we have to use external jar files)

➢ **Steps for connection pooling :-**

a. Create DataSource object

b. Set JDBC parameters like Driver Class Name, Driver URL, DB username, DB password to the DataSource object

c. Set connection pooling properties like connection minimum size, connection maximum size to the DataSource object

d. Get connection object from DataSource

a. Apache Commons DBCP (Database Connection Pooling)

Jar files used are:-

-> mysql-connector.jar

-> commons-pool2-2.8.1.jar

-> commons-pool2-2.2.0.jar

-> commons-logging-1.2.jar

Steps :-

-> BasicDataSource bds=new BasicDataSource();

-> bds.setUrl("jdbc:mysql://localhost:3306/gmail");

bds.setUsername("root");

bds.setPassword("root");

-> bds.setInitialSize(50);

bds.setMaxTotal(100);

-> Connection con=bds.getConnection();

b. Tomcat JDBC

Jar files used are :-

-> tomcat-jdbc.jar

-> tomcat-juli-8.5.4.jar

-> mysql-connector.jar

Steps :-

-> PoolProperties pp=new PoolProperties();

->
pp.setDriverClassName("com.mysql.jdbc.Driver");

pp.setUrl("jdbc:mysql://localhost:3306/gmail");

pp.setUsername("root");

pp.setPassword("root");

-> pp.setMaxActive(100);

pp.setInitialSize(50);

-> DataSource ds=new DataSource();
(apache commons, not javax.sql)

ds.setPoolProperties(pp);

-> Connection con=ds.getConnection();

It is more preferred as compared to Apache Commons DBCP

-> Apache Commons JDBC is single threaded and slow

-> Apache Commons JDBC is complex (approx. 60 classes) while tomcat-jdbc jdbc has only 8 classes

-> Support asynchronous connection retrival

-> Support for distributed transaction (XAConnection)

-> Support JDBC interceptors

c. C3P0

Jar files used are

-> c3p0-0.9.5.2.jar

-> mchange-commons-java-0.2.11.jar

-> mysql-connector.jar

Steps:-

-> ComboPooledDataSource cpds=new ComboPooledDataSource();

-> cpds.setDriverClass("com.mysql.jdbc.Driver");

cpds.setJdbcUrl("jdbc:mysql://localhost:3306/gmail");

cpds.setUser("root");

cpds.setPassword("root");

-> cpds.setMinPoolSize(50);

cpds.setMaxPoolSize(100);

-> Connection con=cpds.getConnection();

d. Proxool

Jar files needed :-

-> proxool-0.9.1.jar

-> proxool-cglib-0.9.1.jar

-> common-logging-1.2.jar

Steps:-

-> ProxoolDataSource pds=new ProxoolDataSource();

-> pds.setDriver("com.mysql.jdbc.Driver");

pds.setDriverUrl("jdbc:mysql://localhost:3306/gmail");

pds.setUser("root");

pds.setPassword("root");

-> pds.setMinimumConnectionCount(50);

pds.setMaximumConnectionCount(100);

-> Connection con=pds.getConnection();

e. BoneCP

f. Hikary

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3. Application server provided connection pooling

-> Steps:-

1. Copy paste mysql-connector.jar in apache/lib

2. Edit apache/conf/context.xml file

3. Copy paste below code

```
<Resource name="jdbc/myds"
type="javax.sql.DataSource"
Username="root" Password="root"
DriverClassName="com.mysql.jdbc.Driver"
Url="jdbc:mysql://localhost:3306/gmail" />
```

4. Then we provide code in Servlets

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